

Income Prediction on US Census Data

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Introduction & Context

► Main question:

- The US Census Bureau periodically collects extensive demographic and economic data to inform public policy and resource allocation in the United States.
- We were provided with an anonymized dataset of approximately 300,000 individuals from the US Census, containing a wide range of features (demographic, economic, social).

► Project goals:

- Identify and explain the characteristics most strongly associated with high income.

Exploratory Data Analysis

"If you torture the data long enough, it will confess to anything."

— Ronald Coase, Economist & Nobel Laureate

Overview and key observations

► Data overview

- ~300,000 records, 40+ features (demographic, economic, social)
- Target variable: Annual income $>$ or $\leq$ \$50,000 (binary classification)

► Initial observations

- Mix of categorical and numerical features.
- Many categorical variables, including some with many modalities (up to 51 modalities).
- Strong class imbalance in target variable: only ~6% of individuals earn $>$ \$50k
- Presence of missing values in several features.
- Presence of duplicate rows.

1 Cleaning

- ◆ **Missing values:**

Mark all missing value labels into NaN

The identified labels are: “?”, *Not in universe*, *Not identifiable*, *NA*, *Do not know*

- ◆ **Label some numerical variables as category:**

Following the documentation, several numeric variables are not measures, but represent only codes and must therefore be typed as categorical, the variables are: *detailed_industry_recode*, *detailed_occupation_recode*, *own_business_or_self_employed*, *veterans_benefits*, and *year*.

- ◆ **Duplicates removal**

2 EDA

Distribution analysis and identification of outliers

- Even even if the data is processed, but typos or outliers can be found.
- Analysis of anomalies and scales is good.
- For some variables there are many categories, which can skew the results.

2 EDA

Missing values

- Here is a summary of variables with most missing values percentage.
- To simplify modeling and avoid information leakage, columns with more than 85% missing values can be dropped*.

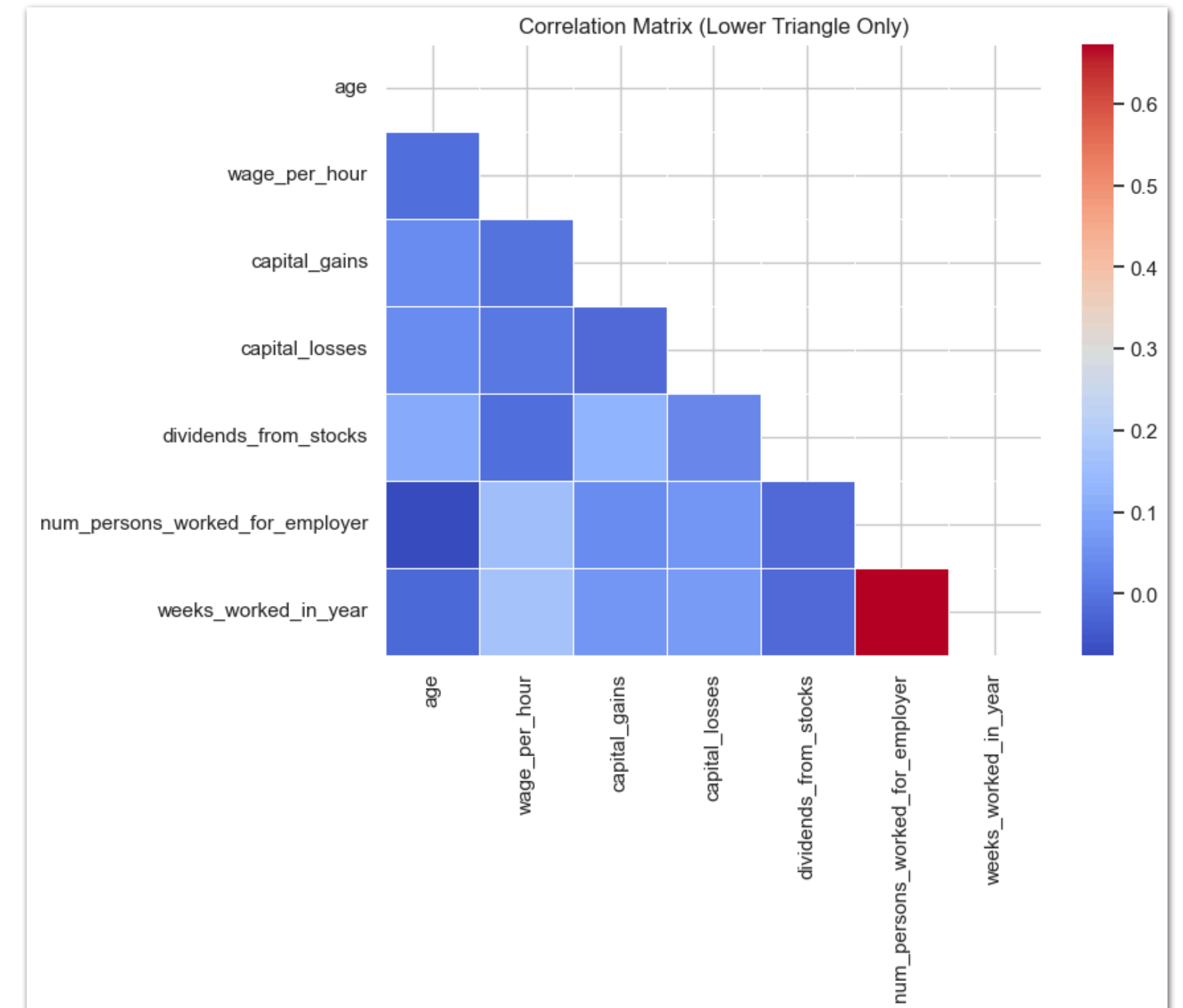
Variable	Missing %
veterans_admin_questionnaire	98.71
reason_for_unemployment	96.07
enroll_in_edu_last_wk	93.03
state_prev_residence	90.38
prev_residence_sunbelt	89.92
region_prev_residence	89.92
family_members_under_18	88.28
member_of_labor_union	87.54
msa_change_code	49.52
reg_change_code	49.25
move_within_reg	49.25
major_occupation_code	35.67
class_of_worker	35.42
country_birth_father	4.18
country_birth_mother	3.80
country_birth_self	2.17
hispanic_origin	0.74

*This decision should always be made **carefully and with domain knowledge**, as some variables may still carry valuable information despite a high proportion of missing values..

2 EDA

Linear correlation analysis: numerical variables

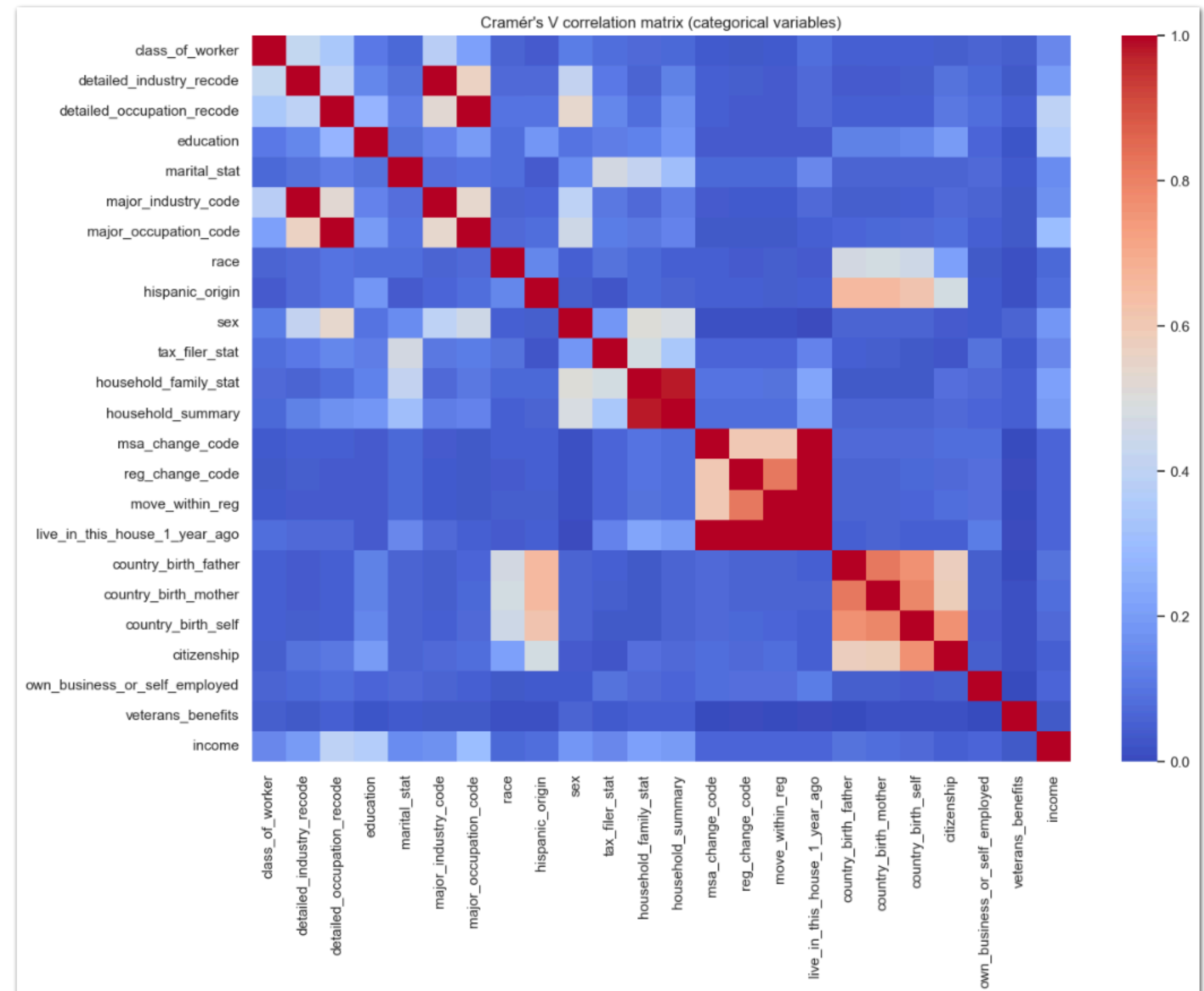
- We examine the linear correlation matrix for all numerical features to detect highly correlated variables, potential redundancy, and to inform future feature selection steps.
- No significant correlations (>0.7) detected.



Data Processing & Engineering

2 EDA Dependency between categorical variables

- We identify categorical variables that are highly correlated with each other to reduce redundancy and improve model performance.
- We use the Cramér's V association matrix, 6 variables were found to be almost duplicative of others (e.g., mobility indicators, detailed industry/occupation codes, and family status), and provides very similar information. The removed variables are : *reg_change_code*, *msa_change_code*, *move_within_reg*, *detailed_industry_recode*, *detailed_occupation_recode*, and *household_family_stat*.



3 Feature engineering & coding

A. Handle missing values:

- We impute missing values according to variable type using simple strategy:
 - Numerical variables: median imputation (robust to extreme values)
 - Categorical variables: impute with a new category "Missing"
- More advanced methods can be applied to better capture relationships between variables (eg. K-Nearest Neighbors, MICE, etc.)

B. Categorical variables encoding:

- For some machine learning models, categorical features can be handled natively (e.g., tree-based models).
- For linear or distance-based models, categorical variables need to be encoded, for example: using one-hot encoding. We can go further and apply more sophisticated techniques such as target encoding, frequency encoding, or embedding representations, depending on the model type and data characteristics.
- Group categories present in less than 1% of all observations under a single category, such as "Other" to prevent overfitting, and improve model generalization*.

*Rare categories may carry meaningful information or have strong relationships with the target variable. Before aggregating them, it is recommended to check their frequency, distribution, and correlation with the outcome.

**Render
jupyter
notebook
with voilà**

Render jupyter notebook with Voilà



Voilà turns Jupyter notebooks into standalone web applications.

<https://github.com/voila-dashboards/voila>

Installation

```
$ pip install voila
```

Add extension to jupyter

```
$ jupyter labextension install @voila-dashboards/jupyterlab-preview
```

Usage examples are available here:

<https://github.com/voila-dashboards/voila/tree/main/notebooks>